INVESTMENT POLICY STATEMENT

Factor-Tilted Technology Strategy

Technology Equities and Algorithmic Trading in Index Futures and Cash
Benchmark: Fidelity Select Technology Portfolio (FSPTX)

1. Purpose and Scope

This Investment Policy Statement (IPS) establishes the objectives, methodology, constraints, and governance framework for a semester-long research portfolio. The portfolio combines a quantitatively selected and weighted technology-stock sleeve with an algorithmic trading sleeve that trades equity index futures and cash. The total portfolio is benchmarked against the Fidelity Select Technology Portfolio (FSPTX), a sector mutual fund investing primarily in companies principally engaged in offering, using, or developing products, processes, or services that depend on technological advances and innovation.

The purpose of this document is to formalize, in advance, how the portfolio will be constructed, how positions will be sized, how risk will be managed, and how performance will be judged, so that decisions made throughout the semester can be evaluated against a pre-committed process rather than adjusted after the fact.

2. Investment Objectives

Primary objectives: The Technology Equities Sleeve seeks to outperform FSPTX, while the Algorithmic Trading sleeve seeks to outperform the S&P 500 (ES) and/or Nasdaq-100 (NQ) on a risk-adjusted basis over the semester.

Secondary objectives:

- Demonstrate a disciplined, repeatable investment process combining factor-based equity selection and optimization with systematic index futures trading and policy-based cash allocation.
- Maintain transparent, auditable model inputs, portfolio weights, and futures strategy versions, with decisions and trades documented in accordance with Section 9.
- Serve as a learning vehicle for factor investing, algorithmic trading, portfolio construction, and performance attribution.

3. Investment Universe and Asset Allocation Policy

The portfolio may invest only in the asset classes below, within the stated ranges, which apply to total portfolio value. The Technology Equities are the primary return engine and are selected and weighted using the factor model in Section 4. The Algorithmic Trading provides additional broad equity exposure and enables experimentation in active trading strategies. The Technology Equities will remain fully invested, with cash used only for when algorithmic trading is not taking positions. We define the Technology Universe as the set of stocks from FSPTX.

Strategic asset allocation policy:

1. Technology Equities: 70–100%. Individual technology-sector equities selected by the factor model, benchmarked to FSPTX.
2. Algorithmic Trading: 0–30%. Algorithmic trading in ES and/or NQ index futures under Section 5. The percentage limit applies to combined notional exposure of the futures across all strategy versions. Cash may be invested into Treasury Bills. The percentage of assets allocated toward Algorithmic Trading will be benchmarked to the S&P 500 and Nasdaq-100.

4. Investment Strategy: Technology Equities

The Technology Equities sleeve will be re-scored and rebalanced on a fixed weekly schedule. At each rebalance date:

1. Refresh all available fundamental data for all companies in the Technology Universe.
2. Compare current sleeve percentages to the policy targets in Section 3. If Technology Equities exceeds its limits, reduce the size of the sleeve.
3. Recompute factors for the Technology Equities.

4. Recalibrate the regression model according to 4.2.
5. Perform the portfolio optimization in 4.3.
6. Compare the new target weights to the current weights and determine the trades required to rebalance. Record the trades for tracking purposes; actual trading will be simulated.
7. Log the rebalance in the portfolio's trade/decision journal.

4.1 Factor Selection

Six factors are used to estimate each company's expected return. They cover valuation, growth, earnings news, price trend, and risk, and each can be computed from prices or analyst consensus estimates at every weekly rebalance date.

Factor	Definition	Expected sign
Forward earnings yield	Next Twelve Months (NTM) consensus Earnings Per Share (EPS) / current price	Positive
Revenue growth forecast	Consensus NTM revenue growth	Positive
EPS growth forecast	Consensus NTM EPS growth	Positive
Momentum	12-month total return excluding the most recent month	Positive
Estimate revisions	3-month % change in NTM consensus EPS	Positive
Volatility	Std. deviation of daily returns, trailing 252 trading days	Negative

The direction of each factor is its expected relationship with future returns, stated in advance and tested by the regression in 4.2. The factor set and directions are fixed for the semester and may change only through the amendment process in Section 10.

4.2 Expected Return Model

Each factor is standardized cross-sectionally at the rebalance date (z-score, capped at ±3). Where a factor is undefined for a company, its z-score is set to 0 (the universe average) so the company is not excluded. Expected next-month excess return is then estimated as:

$$E[R_{shrunk}] = \phi \cdot E[R_{MA}] + (1 - \phi)E\left[R_{factor}\right]$$

where

$$E\left[R_{factor,i}\right] = \alpha + \beta_1 Z_{i,1} + \beta_2 Z_{i,2} + \cdots + \beta_6 Z_{i,6}$$

$$E[R_{MA,i,t}] = \frac{1}{N}\sum_{k=1}^{N}(r_{i,t-k} - r_{f,t-k})$$

The coefficients (β_k) are the average of monthly cross-sectional regression estimates of next-month excess returns on the standardized factors over the trailing 2-year window, using point-in-time data. A coefficient whose sign contradicts the expected sign in 4.1 is set to 0. The set of factors is fixed and can only be revised through amendments. $r_{i,t}$ is the measured return of stock i on month t , and $r_{f,t}$ is the risk-free rate on month t . $N = 18$ is the number of months in the moving average. $\phi = 0.5$ blends the moving average and factor excess return estimates, thereby grounding the noise-prone factor model to a stable estimate of return. N and ϕ can be changed only through amendments.

4.3 Mean Variance Optimization

Sleeve weights are the solution to the following constrained optimization problem, solved at every rebalance date:

$$\max_{w} U(w) = w^T E[R] - \frac{\lambda}{2} w^T C w - \tau \cdot \Sigma |w_i - w_{i,prev}|$$

Decision variables: w : the weights of each equity in the Technology Equity sleeve.

Inputs: $E[R]$ is the vector of expected excess returns from 4.2. C is the covariance matrix of daily returns over the trailing 252 trading days with diagonal shrinkage. λ is the risk-aversion parameter, calibrated at each rebalance date such that the portfolio's ex-ante annualized volatility is expected to be 100% of FSPTX's trailing 252-day annualized volatility. The resulting λ is recorded in the rebalance log for reproducibility. τ is the trading-cost and turnover penalty, set to 15 basis points per unit of turnover, and w_{prev} is the weight vector before rebalancing.

Objective: The portfolio's objective is mean-variance utility, $U(w)$, as defined above. The resulting portfolio's Sharpe ratio, volatility, and tracking error versus FSPTX are recorded at each rebalance.

Additional Constraints on the Technology Equities sleeve:

- Long-only and unlevered; weights sum to 100% of the Technology Equities.
- Single-company cap of 8% of the portfolio.
- Global Industry Classification Standard (GICS) sub-industry cap of 40% (for example, semiconductors, software, hardware), to control the sector concentration inherent in a technology-only universe.
- Ex-ante annualized volatility no greater than 110% of FSPTX's trailing 252-day annualized volatility, and ex-ante tracking error versus FSPTX no greater than 6%.
- Trades are recorded only if a position's weight differs from the optimizer target by more than 1.5 percentage points, to limit turnover.

The problem is a convex quadratic program and can be solved in Excel Solver or Python (for example, scipy or cvxpy). The solver, settings, and inputs are logged at each rebalance so the result can be reproduced.

4.4 Data Timing

All inputs are taken as of the close of the business day before each rebalance date. Consensus estimates are taken from Bloomberg (Best consensus estimates), with FactSet Estimates used to cross-check a sample of companies. Historical values used in the regression are point-in-time to avoid look-ahead bias.

5. Investment Strategy: Algorithmic Trading

The Algorithmic Trading sleeve will trade ES and/or NQ using automated strategies based on three market-structure inputs: Opening Range Breakout (ORB), Level 2/order-flow data, and options gamma exposure (GEX). These inputs may be used independently or combined within a single model, with multiple strategy versions tested and implemented as appropriate. In most cases, the ORB will serve as the initial foundation of the model, while Level 2 and gamma exposure provide additional confirmation and entry/exit structure.

The Opening Range Breakout will be defined using the first 30–90 minutes of the New York trading session. The high and low established during this period create the opening range. A break above the range may establish a long directional bias, while a break below the range may establish a short directional bias. The breakout does not necessarily represent an immediate entry; instead, it may determine which direction the strategy is permitted to trade for the remainder of the trading day. The model may then wait for a retracement, liquidity interaction, or confirmation from Level 2 or gamma data before entering a position.

Level 2 and order-flow data will be used to evaluate the quality of buying and selling activity around important price levels. This may include order-flow delta, which measures the difference between aggressive buying at the offer and aggressive selling at the bid, as well as absorption, exhaustion, bid/ask imbalances, and changes in displayed liquidity.

VWAP, or Volume Weighted Average Price (VWAP) may also be incorporated within the Level 2/order-flow framework. Because VWAP weights price by actual transaction volume, it helps identify areas where significant trading activity and executed liquidity have previously occurred. This can be evaluated alongside Level 2 data, which displays resting bids (“passive buyers”) and asks (“passive sellers”) within the orderbook. Price areas of greater liquidity are particularly important for execution because a deeper pool of available orders can allow positions to be entered or exited with less market impact and reduced slippage, the difference between the best ask and best bid at any given time. VWAP may therefore be used as a liquidity reference and/or key retracement level when evaluating potential entries and exits.

Dealers generally seek to remain approximately delta neutral, requiring them to buy or sell the underlying index or futures as option delta changes. Because options are convex instruments, delta changes aggressively non-linearly as price approaches a strike, particularly near expiration. This makes gamma exposure useful for identifying levels where dealer hedging activity may materially affect intraday price movement. The importance of this framework has increased with the growth of 0DTE SPX options, which now represent a substantial portion of total SPX options activity (roughly 59% on SPX).

Attention may be given to the zero-gamma (the estimated, aggregated, futures converted price where the net gamma exposure is roughly zero) where dealer positioning transitions between positive and negative gamma. Price interaction around this area can create a meaningful contest between buyers and sellers and may indicate a shift in the market's intraday volatility or hedging regime. A confirmed break or rejection of the zero-gamma level, supported by Level 2 information such as delta, absorption, or liquidity changes, may provide the strategy with a clearer directional signal and potential trade entry, particularly when aligned with the “directional bias” given by the ORB model.

Each strategy version will contain defined entry, exit, stop-loss, take-profit, and position-sizing rules. These parameters may be optimized through back testing (with both in and out of sample data to avoid overfitting) with the objective of maximizing risk-adjusted returns rather than simply maximizing net profit. Optimization will remain constrained by the broader IPS, including the 0–30% futures allocation limit, portfolio risk controls, and overall strategy review.

6. Risk Management and Constraints

- Sector concentration: The portfolio is predominantly technology stocks and is therefore intentionally concentrated in one GICS sector; this is an accepted characteristic of the strategy.
- Asset-allocation discipline and automated-trading scope: Each asset class must remain within its approved range in Section 3. The program’s initial mandate is U.S. equity-index exposure through ES and NQ. ES and NQ notional exposure is used to value index futures and may not deviate from boundaries.
- Single-position cap: No single company's weight may exceed 8% of the portfolio at rebalance per Section 4.3, to prevent the scoring formula from producing extreme concentration in one name.
- Minimum diversification: The portfolio will hold a minimum of 30 positions at all times across the two sleeves.

- No leverage and no short positions: The strategy is long-only and unlevered.

7. Performance Benchmark and Evaluation

Total portfolio benchmark: FSPTX. The portfolio's risk-adjusted return is measured against FSPTX's over the same period, using the same return frequency and the 3-Month U.S. Treasury bill as the risk-free rate.

Sleeve benchmarks:

- Technology Equities sleeve: FSPTX
- Algorithmic Trading sleeve: S&P 500 (for ES) and Nasdaq-100 (for NQ)

Evaluation metrics to be tracked and reported at semester end:

- Total portfolio return versus the FSPTX Index and each relevant asset-class benchmark
- Volatility
- Risk-adjusted return (Sharpe ratio; others are optional)
- Maximum drawdown
- Attribution: asset allocation, security selection, and benchmark-relative contributions to outperformance or underperformance

8. Roles and Responsibilities

A. Student Manager Roles

The five-person Committee divides responsibilities between research and risk management. Research covers the Technology Equities methodology, automated Algorithmic Trading strategies, and allocation across the two sleeves. Risk responsibilities are divided between trade review and portfolio monitoring.

The Committee is composed of the following roles:

- Chief Investment Officer (CIO): Leads the investment committee ("the Committee"), oversees execution of the IPS strategy, and coordinates the research and risk functions to ensure the Fund adheres to its pre-committed rules and objectives. Ensures each weekly rebalance (Sections 4 and 5) is completed and logged and presents proposed IPS amendments to the instructor under Section 10.
- Research Analyst - Equity: Covers the Technology Equities Sleeve in full. Runs the weekly factor refresh, regression recalibration (4.2), and optimization (4.3), and logs solver settings, inputs, and λ. Reports target weights and trades to the Committee; does not select companies or weights subjectively. Maintains the equity strategy repository (Section 9B). Proposes trade ideas within this sleeve for Committee approval.
- Research Analyst - Macro / Allocation: Monitors the macro backdrop (rates, inflation, growth) and proposes top-down allocation across index futures and cash within Section 3 ranges. Designs and versions the automated ES/NQ strategies under Section 5, including entry, exit, and sizing rules. Maintains the futures strategy repository (Section 9B).
- Risk Analyst – Trade Review: Reviews each proposed rebalance trade for compliance with Sections 3, 4.3, and 5 before the Committee vote. For automated futures, reviews each strategy version and its risk parameters; orders within approved parameters need no separate review. Assists in periodic adherence analysis (Section 9B).
- Risk Analyst - Portfolio Monitoring: Monitors allocation against Section 3 ranges (including combined futures notional) and the Section 4.3 limits: single-company cap, 30-position minimum, GICS cap, volatility and tracking error limits, and no leverage or shorts. Flags drift or breaches to the Committee, supports rebalancing under Section 4, and assists in periodic adherence analysis (Section 9B).

B. Instructor / Faculty Advisor

The Fund is supervised by the course instructor or faculty advisor, who serves in an advisory capacity to the student fund managers and approves any amendments to this IPS. The instructor/faculty advisor may also facilitate periodic check-ins to review the Fund's performance and adherence to this policy.

9. Documentation, Reporting, and Communication

A. Communication

The Committee will communicate on a continued basis through a shared class communication channel (e.g., Webex, Slack, or Teams) and shared cloud storage. A Committee meeting will be held at least bi-weekly. All five committee members are expected to monitor the Fund and communicate with one another regularly between meetings.

B. Documentation and Reporting

A repository of investment strategies, whether implemented or otherwise, will be maintained jointly by the two Research Analysts. Each is expected to track the strategies they have proposed within their coverage area, and both Risk Analysts are expected to assist in conducting periodic analysis of the Fund's adherence to the risk limits in Section 6.

The Committee will report at the end of each class term, detailing the Fund's simulated performance, asset allocations, risk-management efficacy, and strategic outlook. The Fund will also present any proposed amendments to this Investment Policy Statement at that time, including but not limited to modifications to asset allocation policy ranges, expansion into new asset classes, modifications to risk tolerances, modifications to Fund governance, modifications to rebalancing procedures, and modifications to the inactive management period protocol.

10. Review and Amendment

This IPS will be reviewed at the midpoint of the semester and may be formally amended if the team identifies a methodological flaw (e.g., a factor behaving in an unintended way, data availability issues, or an unworkable weighting formula). Any amendment must be dated, documented with a rationale, and approved by the full team/instructor as applicable, so that the research record reflects an intentional, transparent process rather than after-the-fact adjustment to improve reported results.